Stopping Interpolation Topologically

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Agenda

Project Progress

Outlier Detection

Topological Summaries

Statistical Summaries

Functional Dependencies

Future Perspective

Project Progress

Project Progress

Project Plan

Year	1. Year				2. Year				3. Year			
Quartal	1	2	3	4	1	2	3	4	1	2	3	4
Process analysis												
Literature analysis												
Market analysis												
Pilot study												
Evaluation												
Conception of the tool												
Prototype implementation												
Documentation												
Milestones	M1		M2/M3	M4	M5		M6		M7			M8

Research Questions to Answer

- **RQ1:** What are requirements for exploratory data analysis tools in industrial use?
- **RQ2:** What activities have taken place to exploratory data analysis?
- **RQ3:** What are popular commercial tools and what features do they provide?
- **RQ4:** What are gaps for schema inference and can we close them?

Outlier Detection

Outlier Detection

What is outlier detection?

We assume that most of the time series within a database follow some distribution. If a time series diverges from this distribution significantly it is considered as outlier. We need a function, telling us something about this amount:

- A score $f : X \times I \to \mathbb{R}_+$ can be computed directly.
- A score $f \circ g_i : X \times I \to \mathbb{R}_+$ can be computed on overlapping fixed sized windows, where $g_i : X \times I \to U \times J$ for $U \subset X$ and $J \subset I$ and $\{g_i\}_{i \in I}$ covers $X \times I$.

Academic Tools

Name	**Year**	Distinguish attributes	Univariate analysis	Multivariate analysis	Missing values	Outliers	User friendly
DataScope [2]	2018	✓	✓				
Northstar [3]	2018	✓	✓	✓	✓		

Commercial Tools

Name	Year	Distinguish attributes	Univariate analysis	Multivariate analysis	Missing values	Outliers	User friendly
[]	2018	✓	✓				

Topological Summaries

Topological Summaries

Projektplan

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Statistical Summaries

Statistical Summaries

Projektplan

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Functional Dependencies

Functional Dependencies

Projektplan

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Future Perspective

Goals

What we can do:

- What do we need to infer a schema?
- How is schema inference technically feasible?
- What do we need for the best possible classification?
- Which libraries are suitable for the construction of a prototype?
- What do developers need to create an application?

What we can't do:

- A productively usable tool for schema inference.
- Provide a maintainable tool.
- An additional tool for manual rectification.

References I

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